



CSE 489 | PROJECT REPORT

CHOKRO

Verified Recycling and Eco-Rewards Platform

A cross-platform Flutter application for rewarding verified sustainable actions, supporting 3ZERO Greenpreneurs, and managing auditable eco-points.

Course	CSE 489
Name	Al Sabah Arnish
Student ID	23101437
Email	_____

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1. Project Title

Project Title: Chokro - Verified Recycling and Eco-Rewards Platform

2. Project Overview

Chokro is a mobile-first recycling and eco-rewards application developed with Flutter and Firebase. It connects verified waste disposal, sustainable shopping, and donations through one auditable points wallet. The application targets Android and the web from a single Flutter codebase.

The system is designed around three user-facing profiles: 3ZERO Champion, 3ZERO Greenpreneur, and 3ZERO Admin. Champions submit sustainable actions and use earned points; Greenpreneurs list eco-friendly products and manage orders; Admins review uncertain submissions, manage bins and users, configure policy, and observe platform statistics.

A trusted Node.js service performs sensitive verification and wallet changes. This keeps point awards, order settlement, and donation debits outside the client application and provides a stronger security boundary than direct client-side balance updates.

3. Project Objectives

- Encourage correct waste disposal by converting verified sustainable actions into reward points.
- Create an accountable earn-and-spend cycle through a unified wallet and immutable transaction ledger.
- Give local eco-friendly entrepreneurs a dedicated marketplace and order-management channel.
- Provide administrators with review queues, policy controls, and meaningful platform statistics.

4. Project Features

By the end of this course, the following features have been implemented:

- a) **Login system.** Email/password registration, secure sign-in and sign-out, validation, account-state checks, and profile-aware routing are implemented with Firebase Authentication.
- b) **Reporting system.** The application provides administrative dashboards, live platform statistics, pending review queues, user submission histories, order records, wallet ledgers, and appeal outcomes for accountable monitoring.
- c) **Role and profile management.** 3ZERO Admin, 3ZERO Greenpreneur, and 3ZERO Champion workspaces use a clear profile hierarchy. Switching profiles changes the workspace while the stored role remains the authority for protected actions.

- d) **Verified disposal workflow.** A four-step scan, photo, location, and confirmation process collects evidence. Server-side checks evaluate bin distance, photo provenance, duplicate hashes, screening results, and submission limits.
- e) **Eco-points wallet and transaction history.** Approved activity credits are recorded in an immutable ledger. Points can be spent during checkout or donated to selected green initiatives, with idempotent server operations to avoid duplicate charges or rewards.
- f) **Green marketplace and order system.** Greenpreneurs can publish and manage products. Champions can browse and filter the catalogue, maintain a cart, place multi-seller orders, apply points, follow order status, and confirm receipt.
- g) **Donation and payment prototypes.** The system supports point donations and clearly labelled bKash, Nagad, and card payment simulations without collecting credentials or moving real money.
- h) **Administrative review and appeals.** Flagged disposals, self-reported claims, Greenpreneur applications, and user appeals can be reviewed through dedicated admin queues with recorded decisions and responses.
- i) **Bin and notification management.** Admins can register bins and generate printable high-error-correction QR labels. Firebase Cloud Messaging supports decision and account-related notifications.
- j) **Security and validation.** Firestore rules enforce ownership and server-only fields, while the Node service performs trusted decisions, wallet mutations, rate limiting, and cross-record transaction checks.

5. System Design and Technology

Chokro separates presentation, state management, cloud data, and trusted business decisions. This architecture supports mobile and web interfaces while keeping sensitive balance and verification logic on the server.

Layer	Implementation
Client application	Flutter and Dart for Android and web from one codebase
Navigation and state	go_router and Riverpod for routing and reactive state
Identity	Firebase Authentication with email/password accounts
Database	Cloud Firestore with indexes and strict security rules
Trusted service	Node.js 22, Express, and Firebase Admin SDK
Evidence capture	QR scanning, camera/image handling, compression, and geolocation

Layer	Implementation
Media and screening	Cloudinary-backed media handling and optional AI-assisted screening
Notifications	Firebase Cloud Messaging
Quality assurance	Flutter tests, Jest server tests, and Firestore emulator rules tests

5.1 Main Operational Workflow

1. **Account access.** A user registers or signs in and enters the correct profile workspace.
2. **Action capture.** A Champion scans a registered bin, takes a photograph, confirms location, and submits evidence.
3. **Trusted verification.** The server checks identity, location, photo evidence, duplicate risk, policy limits, and screening results.
4. **Decision and reward.** A passing submission receives an atomic wallet credit; uncertainty is routed to human review.
5. **Use of value.** The Champion spends points in the marketplace or donates them to a green initiative.
6. **Monitoring.** Admins review queues, resolve appeals, manage policy and bins, and inspect aggregate statistics.

6. Online Resources Used

The following online materials were used as learning references, implementation guides, and documentation sources during development.

a) Reference

- **W3Schools:** [JavaScript JSON tutorial](#) - background for JSON request and response structures.
- **Official documentation:** [Flutter documentation](#) - widgets, navigation, platform setup, and application development guidance.
- **Official documentation:** [Add Firebase to a Flutter app](#) - FlutterFire and Firebase project configuration.
- **Official documentation:** [Firebase email/password authentication for Flutter](#) - account registration and login behavior.
- **YouTube video 1:** [Firebase x Flutter Tutorial - Firebase Authentication](#) - practical email/password authentication walkthrough.

- **YouTube video 2:** [Flutter Firebase Tutorial for Beginners](#) - introduction to Firebase Authentication and Cloud Firestore with Flutter.

b) Stack Overflow or GitHub Links

- **Project repository:** [alsabaharnish/chokro](#) - source code and project history.
- **GitHub reference:** [firebase/flutterfire](#) - official Firebase plugins and examples for Flutter.
- **GitHub reference:** [firebase/firebase-admin-node](#) - official Firebase Admin SDK for the trusted Node.js service.

7. Future Enhancements

The following enhancements can be added to the current system to improve usability, reliability, reach, and operational value:

- Understanding of system.** Add a first-time guided tour, short role-based tutorials, contextual help, sample disposal demonstrations, and clearer explanations of how points are earned, spent, and donated.
- Login system.** Add email verification, password reset, optional Google sign-in, multi-factor authentication for administrators, active-session management, and clearer recovery paths for locked or suspended accounts.
- Reporting system.** Expand the dashboard with date and location filters, trend charts, bin-level utilization reports, Greenpreneur sales summaries, downloadable PDF/CSV exports, and scheduled administrator reports.
- Production payments.** Replace payment simulations with approved bKash, Nagad, or card-gateway integrations using server-verified callbacks, secure receipts, reconciliation, refunds, and dispute handling.
- Offline support and synchronization.** Allow evidence drafts to be captured in areas with weak connectivity and uploaded later with freshness checks, conflict handling, and visible sync status.
- Smarter verification.** Improve image-quality guidance, duplicate detection, risk scoring, reviewer explanations, and model monitoring while retaining a fail-to-review policy.
- Accessibility and localization.** Add full Bangla localization, larger-text support, screen-reader improvements, high-contrast modes, and simplified workflows for first-time smartphone users.
- Production scaling and operations.** Introduce centralized logging, audit dashboards, automated backups, performance monitoring, stricter abuse controls, penetration testing, and a documented incident-response process before public deployment.

8. Conclusion

Chokro demonstrates a complete sustainability reward cycle: users can verify an eco-friendly action, receive points through a trusted decision process, and reuse those points through marketplace purchases or initiative donations. The project also includes role-based administration, evidence review, appeals, reporting, and security controls appropriate for an academic prototype.

The implemented system satisfies the central CSE 489 goal of delivering a working mobile application while providing a clear path toward stronger onboarding, authentication, analytics, production payments, accessibility, and operational scale.